CHEMO
DRAWING U403
UNIVERSAL FARMA
MATERIAL TYPE LEAFLET
SCALE 100 %
DIMENSIONS 975 x 495 (66x45) ± 1 mm
MATERIAL # / PRODUCT : ******
VERSION : **
DATE : ******
SUBSTITUTES MAT. # : ******
PANTONES
Only a max. of 6 printing colors / Please, indicate below the colours to be used
LINE 1: FRONT SIDE PRINTING COLOUR
LINE 2: BACK SIDE PRINTING COLOUR
LINE 3: TECH. COLOUR NO PRINTING COLOUR
DIE CUT
FINAL FOLDED
HEAT SEALED
GLUE POINT
FOR DIE FOLDED
CAUSE:
LAUNCH
CUSTOMER CHANGE
INTERNAL CHANGE
LAETUS : *****
FOLDED
NO
YES
FOLDED DIMENSIONS: fold to 6 x 45 mm
ONLY FOR POSITION
Customer shall be responsible for the approval of the approved artwork and the limited material reflected in this drawing. Therefore, Customer shall blue any and all cost and/or expense, and it shall keep Universal Farma harmless from any claim, cost and/or expenses related to the correctness and accuracy of such information.
DRAWED BY:
XAVIER MONTERDE TECHNICIAN
APPROVED BY CUSTOMER :
(REGULATORY AFFAIR OR QUALITY ASSURANCE) / SIGNATURE & DATE.
With the signed proof of this file you commit to have had reviewed and agreed with the following points as part of your review:
- Keyline used: Free varnish areas / Colors / Text.
- Universal Farma is not responsible for any legal issues related to clearly communicate this change to Chemo's artworks department, otherwise Chemo will not be responsible.
Any other info included in the file.
PLEASE, MAKE SURE THE IMAGES HAVE A RESOLUTION AT LEAST 300 PP
PLEASE, CHECK CAREFULLY ANY LAYERS FOR THE CORRECT ELEMENTS
FRONT SIDE / PLEASE, INCLUDE YOUR DESIGN ON THIS KEYLINE
150 mm
245 mm
495 mm
AVAMAB
Bevacizumab
Solution For Infusion
KALBE
AVAMAB
Bevacizumab
Solution For Infusion
KALBE
LISTA CONTROL DE IMPRESIÓN (FINAL FOLDED INCLUDED)
66 mm
904 mm
975 mm

AVAMAB

AVAMAB is a biosimilar to AVASTIN.

This medicinal product is subject to additional monitoring. This will allow quick identification of new safety information. Healthcare professionals are asked to report any suspected adverse reactions. See section **Undesirable Effects** for how to report adverse reactions.

Composition

Each mL of concentrate contains 25 mg of bevacizumab*.
Each 4 mL vial contains 100 mg of bevacizumab.
Each 16 mL vial contains 400 mg of bevacizumab.
For dilution and other handling recommendations, see section **Special precautions for disposal and other handling.**

*Bevacizumab is a recombinant humanised monoclonal antibody produced by DNA technology in Chinese Hamster Ovary cells.
Active ingredient: bevacizumab 25mg/ml

List of excipients:

Trehalose dihydrate
Monobasic sodium phosphate monohydrate
Disodium phosphate
Polysorbate 20
Water for injections

Pharmaceutical Form

Concentrate for solution for infusion (sterile concentrate). Colourless to yellowish or brownish liquid with opalescence.

Pharmacological Properties

Pharmacodynamic Properties

Mechanism of action

Bevacizumab is a recombinant humanised monoclonal antibody that selectively binds to and neutralises the biologic activity of human vascular endothelial growth factor (VEGF). Bevacizumab contains human framework regions with antigen binding regions of a humanised murine antibody that binds to VEGF. Bevacizumab is produced by CHO cells in a nutrient medium containing gentamicin which is not present in the final formulation. This process was verified with respect to Asavab.
Approximately 350 patients were randomized into each of the 4 study arms in the Part II of the trial.

Table 2 Treatment Regimens in Study NO16966 (mCRC)

Treatment	Starting dose	Schedule
FOLFOX-4 or FOLFOX-4 + bevacizumab	Oxaliplatin 85 mg/m² intravenous 2 h	Oxaliplatin on day 1
	Leucovorin 200 mg/m² intravenous 2 h	Leucovorin on day 1 and 2
	5-fluorouracil intravenous bolus 400 mg/m² intravenous bolus, 600 mg/m² intravenous 22 h	5-fluorouracil intravenous bolus/infusion, each on days 1 and 2
	Placebo or bevacizumab 5 mg/kg intravenous 30-90 min	Day 1, prior to FOLFOX-4, every 2 weeks
XELOX or XELOX + bevacizumab	Oxaliplatin 130 mg/m² intravenous 2 h	Oxaliplatin on day 1
	Capecitabine 1,000 mg/m² oral bid	Capecitabine oral bid for 2 weeks (followed by 1 week off treatment)
	Placebo or bevacizumab 7.5 mg/kg intravenous 30-90 min	Day 1, prior to XELOX, q 3 weeks

5-Fluorouracil: intravenous bolus injection immediately after leucovorin

The primary efficacy parameter of the trial was the duration of progression-free survival. In this study, there were two primary objectives: to show that XELOX was non-inferior to FOLFOX-4 and to show that Avastin in combination with FOLFOX or XELOX chemotherapy was superior to chemotherapy alone.
Both co-primary objectives were met:

1 Non-inferiority of the XELOX-containing arms compared with the FOLFOX-4- containing arms in the overall comparison was demonstrated in terms of progression-free survival and overall survival in the eligible per-protocol population

II Superiority of the Avastin -containing arms versus the chemotherapy alone arms in the overall comparison was demonstrated in terms of progression-free survival in the ITT population (Table 3). Secondary PFS analyses, based on Independent Review Committee (IRC)- and 'on-treatment'-based response assessments, confirmed the significantly superior clinical benefit for patients treated with Avastin (subgroup analyses shown in Table 3), consistent with the statistically significant benefit observed in the pooled analysis.

Table 3 Key efficacy results for the superiority analysis (ITT population, Study NO16966)

Endpoint (months)	FOLFOX-4 or XELOX + placebo (n = 701)	FOLFOX-4 or XELOX + Avastin (n = 699)	p- value
Primary endpoint			
Median PFS*	8.0	9.4	0.0023
Hazard ratio (97.5% CI)*	0.83 (0.72 - 0.95)		
Secondary endpoints			
Median PFS (on treatment)**	7.9	10.4	< 0.0001
Hazard ratio (97.5% CI)	0.63 (0.52 - 0.75)		
Median FPS (Indep.review)**	8.5%	11%	< 0.0001
Hazard ratio (97.5% CI)	0.70 (0.58-0.83)		
Overall response rate (invest. assessment)**	49.2%	46.5%	
Overall response rate (indept. Review)**	37.5%	37.5%	
Median overall survival*	19.9	21.2	0.0769
Hazard ratio (97.5% CI)	0.89 (0.76 - 1.03)		

* Overall survival analysis at clinical cut-off 31 January 2007
**Primary analysis at clinical cut-off 31 January 2006
* relative to control arm

ECOG E3200

This was a phase III randomised, active-controlled, open-label study investigating Avastin 10 mg/kg in combination with leucovorin with 5-fluorouracil bolus and then 5-fluorouracil infusional, with iv oxaliplatin (FOLFOX-4), administered in a 2-weekly schedule in previously treated patients (second line) with advanced colorectal cancer. In the chemotherapy arms, the FOLFOX-4 regimen used the same doses and schedule as shown in Table 4 for Study NO16966.
The primary efficacy parameter of the trial was overall survival, defined as the time from randomization to death from any cause. A total of 829 patients (with twenty-nine patients characterized (292 FOLFOX-4, 293 Avastin + FOLFOX-4 and 244 Avastin monotherapy). The addition of Avastin to FOLFOX-4 resulted in a statistically significant prolongation of survival. Statistically significant improvements in progression-free survival and objective response rate were also observed (see Table 4).

Table 4 Efficacy Results for Study E3200

	E3200		
	FOLFOX-4	FOLFOX-4 + Avastin*	
Number of patients	292	293	
Overall survival			
Median (months)	10.8	13.0	
95% CI	10.12 - 11.86	12.09 - 14.03	
Hazard ratio*	0.751 (p-value = 0.0012)		
Progression-free survival			
Median (months)	4.5	7.5	
Hazard ratio	0.518 (p-value < 0.0001)		
Objective response rate	8.6%	22.2% (p-value < 0.0001)	

* 10 mg/kg every 2 weeks.
* Relative to control arm.

No significant difference was observed in the duration of overall survival between patients who received Avastin monotherapy compared to patients treated with FOLFOX-4. Progression-free survival and objective response rate were inferior in the Avastin monotherapy arm compared to the FOLFOX-4 arm.

ML18147

This was a Phase III randomized, controlled, open-label trial investigating Avastin 5.0 mg/kg every 2 weeks or 7.5 mg/kg every 3 weeks in combination with fluoropyrimidine-based chemotherapy versus fluoropyrimidine-based chemotherapy alone in patients with metastatic colorectal cancer who have progressed on a first-line Avastin containing regimen. Patients with histologically confirmed mCRC and disease progression were randomized 1:1 within 3 months after discontinuation of Avastin first-line therapy to receive fluoropyrimidine/oxaliplatin or fluoropyrimidine/irinotecan-based chemotherapy (chemotherapy determined depending on first-line chemotherapy) with or without Avastin. Treatment was given until progressive disease or unacceptable toxicity. The primary outcome measure was overall survival (OS) defined as the time from randomization until death from any cause.

A total of 820 patients were randomized. The addition of Avastin to fluoropyrimidine-based chemotherapy resulted in a statistically significant prolongation of survival in patients with metastatic colorectal cancer who have progressed on a first-line Avastin -containing regimen (ITT = 819) (see Table 5).

Table 5 Efficacy Results for Study ML18147

	ML18147	
	Fluoropyrimidine/irinotecan or Fluoropyrimidine/oxaliplatin based chemotherapy	Fluoropyrimidine/irinotecan or Fluoropyrimidine/oxaliplatin based chemotherapy + Avastin*
Number of patients	410	409
Overall survival		
Median (months)	9.8	11.2
95% confidence interval	9-11	10-12
Hazard ratio (95% confidence interval)	0.81 (p-value = 0.0062)	
Progression-free survival		
Median (months)	4.1	5.7
Hazard ratio (95% confidence interval)	0.68 (p-value < 0.0001)	
Objective Response Rate (ORR)		
Rate	3.9%	5.4% (p-value = 0.3113)

* 2.5 mg/kg/week
Statistically significant improvements in progression-free survival were also observed.
Objective response rate was low in both treatment arms and did not meet statistical significance.

Adjuvant Colon Cancer (aCC)

BO17920

This was a phase III randomized open-label, 3-arm study evaluating the efficacy and safety of Avastin administered at a dose equivalent to 2.5 mg/kg/week or either a 2-weekly schedule in combination with FOLFOX4, or an 3- weekly schedule in combination with XELOX versus FOLFOX4 alone as adjuvant chemotherapy in 3457 patients with high-risk stage II and stage III colon cancer.
More relapses and deaths due to disease progression were observed in both Avastin arms compared to the control arm. The primary objective of prolonging disease free survival (DFS) in patients with stage III colon cancer (n = 2867) by adding Avastin to either chemotherapy regimen was not met. The hazard ratios for DFS were 1.17 (95% CI: 0.98-1.39) for theFOLFOX + Avastin arm and 1.07 (95% CI: 0.90-1.28) for the XELOX + Avastin arm.

Metastatic breast cancer (mBC)

ECOG E2100

E2100 was an open-label, randomized, active-controlled, controlled multicentre clinical trial evaluating Avastin in combination with paclitaxel versus paclitaxel monotherapy as first-line treatment for metastatic breast cancer patients who had not previously received chemotherapy for locally recurrent and metastatic disease. Prior hormonal therapy for the treatment of metastatic disease was allowed. Adjuvant taxane therapy was permitted only if it was completed at least 12 months prior to study entry.
Patients were randomized to paclitaxel 90 (mg/m²PPP 2PPP IV over 1 hour once weekly for three out of four weeks) or in combination with Avastin (10 mg/kg IV infusion every two weeks). Patients were to continue assigned study treatment until disease progression. In cases where patients discontinued chemotherapy prematurely, treatment with Avastin as a single agent was continued until disease progression. The primary endpoint was progression free survival (PFS), as assessed by investigators. In addition, an independent review of the primary endpoint was also conducted. Of the 722 patients in the study, the majority of patients (80%) had HER2-negative disease. A small number of patients (~ 20%) had HER-2 receptor status that was either unknown (8%) or positive (2%). Patients who were HER-2 positive had either received previous treatment with trastuzumab or were considered unsuitable for trastuzumab. The majority (65%) of patients had received adjuvant chemotherapy including 19% who had prior taxanes and 49% who had prior anthracyclines. The patient characteristics were similar between the study arms. The results of this study are presented in Table 6.

Table 6 Study E2100 Efficacy Results: Eligible Patients

Progression-free survival				
	Investigator assessment*		IRF assessment	
	Paclitaxel (n = 354)	Paclitaxel/ Avastin (n = 354)	Paclitaxel (n = 354)	Paclitaxel/ Avastin (n = 368)
Median PFS (months)	5.8	11.4	5.8	11.3
HR (95% CI)	0.421 (0.343, 0.516)		0.483 (0.385, 0.607)	
p-value	< 0.0001		< 0.0001	
Response rates (for patients with measurable disease)				
	Investigator assessment		IRF assessment	
	Paclitaxel (n = 273)	Paclitaxel/ bevacizumab (n = 252)	Paclitaxel (n = 243)	Paclitaxel/ bevacizumab (n = 229)
% pts with objective response	23.4	48.0	22.2	49.8
p-value	< 0.0001		< 0.0001	

* Primary analysis

Congestive heart failure (CHF) (see section Undesirable Effects)

Events consistent with CHF were reported in clinical trials. The findings ranged from asymptomatic declines in left ventricular ejection fraction to symptomatic CHF, requiring treatment or hospitalisation. Caution should be exercised when treating patients with clinically significant cardiovascular disease such as pre-existing coronary artery disease, or congestive heart failure with Bevacizumab. Most of the patients who experienced CHF had metastatic breast cancer and had received previous treatment with anthracyclines, prior radiotherapy to the left chest wall or other risk factors for CHF were present.

Neutropenia and infections (see section Undesirable Effects)

Increased rates of CTCAE grade 3 neutropenia, febrile neutropenia, or infection with or withoutneutre neutropenia (including some fatalities) have been observed in patients treated with some myelotoxic chemotherapy regimens plus bevacizumab in comparison to chemotherapyalone.

Wound healing (see section Undesirable Effects)

Bevacizumab may adversely affect the wound healing process. Serious wound healing complications, with a fatal outcome have been reported. Bevacizumab therapy should not be initiated for at least 28 days following major surgery or until the surgical wound is fully healed. Bevacizumab should be withheld until healing complications during therapy, treatmentshould be withheld until the wound is fully healed. Bevacizumab therapy should be withheld for elective surgery. Patients, including fatal cases, has rarely been reported in patients treated with bevacizumab. This condition is usually secondary to wound healing complications, gastrointestinal perforation orfistula formation. Bevacizumab therapy should be discontinued in patients who develop necrotising fasciitis, and appropriate treatment should be promptly initiated.

Proteinuria (see section Undesirable Effects)

In clinical trials, the incidence of proteinuria was higher in patients receiving bevacizumab in combination with chemotherapy compared to patients treated with chemotherapy alone. Grade 4 proteinuria (nephrotic syndrome) was seen in up to 1.4% of patients treated with bevacizumab. In the event of nephrotic syndrome, bevacizumabshould be discontinued.

Hypersensitivity reactions/infusion reactions (see section Undesirable Effects)

Patients may be at risk of developing infusion/hypersensitivity reactions. Close observation of the patient during and following the administration of bevacizumab is recommended for at least one hour.In infusion of a therapeutic humanised monoclonal antibody. If a reaction occurs, the infusion should be discontinued and appropriate medical therapies should be administered. A systematic premedication is not warranted.

Ovarian failure/fertility

Bevacizumab may impair female fertility (see sections Fertility, Pregnancy and Lactation and Undesirable Effects). Therefore fertility preservation strategies should be discussed with women of childbearing potential before treatment with bevacizumab.

Interaction with Other Medicinal Products and Other Forms of Interaction

Effect of antineoplastic agents on bevacizumab pharmacokinetics No clinically relevant pharmacokinetic interaction of co-administered chemotherapy on bevacizumab pharmacokinetics has been observed based on the results of a population PK analysis. There was neither statistical significance nor clinically relevant difference in clearance of Avastin in patients receiving Avastin monotherapy compared to patients receiving Avastin in combination with interferon-alfa 2a or other chemotherapies (IFL, 5-FU/LV, carboplatin/paclitaxel, capecitabine, docetaxel, oxaliplatin/5-FU/LV).

Effect of bevacizumab on the pharmacokinetics of other antineoplastic agents

No clinically relevant interaction of bevacizumab was observed on the pharmacokinetics of coadministered interferon-alfa, or the chemotherapies irinotecan (and its active metabolite SN38), capecitabine, oxaliplatin (as determined by measurement of free and total platinum), and cisplatin. Conclusions on the impact of bevacizumab on pharmacokinetic parameters of chemotherapies cannot be drawn.

Combination of bevacizumab and sunitinib malate

In two clinical trials of metastatic renal cell carcinoma, microangiopathic haemolytic anaemia (MAHA) was reported in 7 of 91 patients receiving bevacizumab 10 mg/kg every two weeks) and sunitinib malate (50 mg daily) combination.
MAHA is a haemolytic disorder which can present with red cell fragmentation, anaemia, and thrombocytopenia. In addition, hypertension (including hypertensive crisis), elevated creatinine, and neurological symptoms were observed in some patients. All these findings were reversible upon discontinuation of bevacizumab and sunitinib malate (see Hypertension, Proteinuria, PRES in section Special Warnings and Precautions for Use).

Radiotherapy

The safety and efficacy of concomitant administration of radiotherapy and bevacizumab hasnot been established.

Fertility, Pregnancy and Lactation

Women of childbearing potential/Contraception

In women with childbearing potential, appropriate contraceptive measures should be used during Bevacizumab therapy. Based on pharmacokinetic considerations, contraceptive measures should be used for at least 6 months following the last dose of Bevacizumab

Pregnancy

Angiogenesis has been shown to be critically important to foetal development. The inhibition of angiogenesis following administration of Bevacizumab could result in an adverse outcome of pregnancy. There are no adequate and well-controlled studies in pregnant women. IgGs are known to cross the placental barrier, and Bevacizumab may therefore cause impairment in the foetus. In the post-marketing setting, cases of foetal anomalies in women treated with bevacizumab alone or in combination with known embryotoxic chemotherapeutics have been observed. Therefore, Bevacizumab should not be used during pregnancy.

Lactation

It is not known whether bevacizumab is excreted in human milk. As maternal IgG is excreted in milk and bevacizumab could impair growth and development (see section Preclinical Safety Data), women must discontinue breast-feeding during therapy and not breast-feed for at least six months following the last dose of bevacizumab.

Fertility

Bevacizumab may impair female fertility. Women of child-bearing potential should beadvised of fertility preservation strategies prior to starting treatment with bevacizumab. Repeat dose safety studies in animals have shown that bevacizumab may have an adverseeffect on female fertility. A subsitudy with 295 premenopausal women showed a higher incidence of new cases of ovarian failure in the bevacizumab group compared to control group. After discontinuation of bevacizumab treatment, ovarian function recovered in the majority of patients. Long term effects of the treatment with bevacizumab on the fertilityare unknown.

Undesirable Effects

Clinical Trials

Summary of safety profile

Clinical trials have been conducted in patients with various malignancies treated with bevacizumab, predominantly in combination with chemotherapy. The safety profile from a clinical trial population of approximately 5,500 patients is presented in this section. For post marketing experience see section Postmarketing below. See section Clinical Efficacy Studies for details of major study designs, including study designs and major efficacy results. The most serious adverse events were:

- Gastrointestinal perforations (see section Warnings and Precautions, General)
- Haemorrhage including pulmonary haemorrhage/haemoptysis, which is more common in NSCLC patients (see section Warnings and Precautions, General)
- Arterial thromboembolism (see section Warnings and Precautions, General) Analyses of the clinical safety data suggest that the occurrence of hypertension andproteinuria with bevacizumab are likely to be dose-dependent.

The most frequently observed adverse drug reactions across clinical trials in patients receiving bevacizumab were hypertension, fatigue or asthenia, diarrhoea and abdominal pain

Tabulated list of adverse reactions

Table 17 lists adverse drug reactions associated with the use of bevacizumab in combination with different chemotherapy regimens in multiple indications, by MedDRA system organ class. The corresponding frequency category for each adverse drug reaction is based on the following convention: very common (≥ 1/10); common (≥1/100 to <1/10); uncommon (≥ 1/1,000 to <1/100); rare (≥1/10,000 to <1/1,000); very rare (<1/10,000). These reactions had occurred either with at least a 2% difference compared to the control arm [NCI-CTC [common toxicity criteria] Grade 3-5 reactions) or with at least a 10% difference compared to the control arm (NCI-CTC Grade 1-5 reactions), in at least one of the major clinical trials. Adverse drug reactions are added to the appropriate category in the table below according to the highest incidence seen in any of the major clinical trials. Within each frequency grouping adverse drug reactions are presented in order of decreasing seriousness. Some of the adverse reactions are reactions commonly seen with chemotherapy, however, bevacizumab may exacerbate these reactions when combined with chemotherapeutic agents. Examples include palmar-plantar erythrodysaesthesia syndrome with pegylated liposomal doxorubicin or capecitabine, peripheral sensory neuropathy with paclitaxel or oxaliplatin, and nail disorders or alopecia with paclitaxel.

Table 7 Efficacy results for study E4599

	Arm 1 Carboplatin/Paclitaxel	Arm 2 Carboplatin/ Paclitaxel + Avastin 15 mg/kg q 3 weeks
Number of patients	444	434
Overall survival		
Median (months)	10.3	12.3
Hazard ratio	0.80 (p = 0.003) 95% CI (0.69; 0.93)	
Progression-free survival		
Median (months)	4.8	6.4
Hazard ratio	0.65 (p < 0.0001) 95% CI (0.56; 0.76)	
Overall response rate		
Rate (percent)	12.9	29.0 (p < 0.0001)

Table 8 Efficacy results for trial BO17704

	Cisplatin/Gemcitabine + placebo	Cisplatin/Gemcitabine + Avastin 7.5 mg/kg q 3 weeks	Cisplatin/Gemcitabine Avastin 15 mg/kg q 3 weeks
Number of patients	347	345	351
Progression-free survival			
Median (months)	6.1	6.7 (p = 0.0226)	6.5 (p = 0.0301)
Hazard ratio		0.75 [0.62; 0.91]	0.82 [0.68; 0.98]
Best overall response rate*	20.1%	34.1% (p < 0.0001)	30.4% (p = 0.0023)

* patients with measurable disease at baseline

Overall survival			
Median (months)	13.1	13.6 (p = 0.4203)	13.4 (p = 0.7613)
Hazard ratio		0.93 [0.78; 1.11]	1.03 [0.86; 1.23]

Advanced and/or metastatic renal cell cancer (mRCC)
BO17705

Study BO17705 was a multicenter randomised, double-blind phase III trial conducted to evaluate the efficacy and safety of Avastin in combination with interferon (IFN)-alfa-2a (Roferon®) versus IFN-alfa-2a alone as first-line treatment in mRCC. The 649 randomized patients (641 treated) had clear cell mRCC, Karnofsky Performance Status (KPS) of ≥70%, no CNS metastases and adequate organ function. IFN-alfa-2a (s)Kweek at a recommended dose of 9 MIU plus Avastin (10 mg/kg q2w) or placebo was given until disease progression. Patients were stratified according to country and Motzer score at the treatment arms were shown to be well balanced for the prognostic factors.
The primary endpoint was overall survival, with secondary endpoints for the study including progression-free survival. The duration of Avastin to IFN-alfa-2a significantly increased PFS and produced a 28% increase in response rate. These results have been confirmed through an independent radiological review. However, the increase in PFS and response rate did not translate into improved survival for 2 months was not significant (HR = 0.91). A high proportion of patients (approximately 63% IFN/placebo; 55% Avastin IFN) received a variety of non-specified, post-protocol anti-cancer therapies, including subsequent agents, which may have impacted the analysis of overall survival.
The efficacy results are presented in Table 9.

Table 9. Efficacy results for trial BO17705

	BO17705		
	Placebo + IFN a [1] (n = 322)	IFN a + Avastin [1] (n = 327)	
Number of patients			
Progression-free survival			
Median (months)	5.4	10.2	
Hazard ratio 95% CI	0.63 0.52; 0.75 (p-value < 0.0001)		
Objective response rate (%) in patients with measurable disease			
N	289	306	
Response rate	12.8%	31.4% (p-value < 0.0001)	
Overall survival			
Median (months)	21.3	23.3	
Hazard ratio 95% CI	0.91 0.76; 1.10 (p-value 0.3360)		

An exploratory multivariate Cox regression model using backward selection indicated, that the following baseline prognostic factors were strongly associated with survival independent of treatment: serum lactate dehydrogenase level, calcium level, platelet count, baseline haemoglobin level and time from diagnosis to treatment. Adjustment for these covariates did not change the estimated treatment hazard ratio of 0.78 (95% CI [0.63; 0.96], p = 0.0219), indicating a 22% reduction in the risk of death for patients in the Avastin + IFN-alfa-2a arm compared to IFN-alfa-2a alone.
Ninety-seven (97) patients in the IFN alfa-2a arm and 131 patients in the Avastin armreduced the dose of IFN-alfa-2a from 9 MIU to 4.5 MIU or 3 MIU, three times a week as pre-specified in the protocol. Dose reduction of IFN alfa-2a did not appear to affect the efficacy of the combination of Avastin and IFN alfa-2a, based on PFS event free rates over time, as shown by a sub-group analysis. The 131 patients in the Avastin + IFN alfa-2a arm who reduced and maintained the IFN alfa-2a dose at 6 or 3 MIU during the study, exhibited at 6, 12 and 18 months, PFS event free rates of 73, 52 and 21% respectively, as compared to 61, 43 and 17% in the total population of patients receiving Avastin + IFN alfa-2a.

AVF2938

This was a randomized, double-blind, phase II clinical study investigating Avastin 10 mg/kg in a 2 weekly schedule with the same dose of Avastin in combination with 150 mg dalcetrolimib, in patients with metastatic clear cell RCC. A total of 104 patients were randomized to treatment in this study, 53 to Avastin 10 mg/kg q2w plus placebo and 51 to Avastin 10 mg/kg q2w plus erlotinib 150 mg daily. The analysis of the primary endpoint showed no difference between the Avastin + Pl arm and the Avastin + Erl arm (median PFS 8.5 versus 9.9 months). Seven patients in each arm had an objective response.

Glioblastoma

AVF3708g

The efficacy and safety of Avastin as treatment for patients with glioblastoma was studied in an open-label, multicentre, randomized, non-comparative study (study AVF3708g).
Glioblastoma patients in first or second relapse after prior radiotherapy (completed at least 8 weeks prior to recruitment) and temozolomide (if appropriate) were randomized (1:1) to receive Avastin (10 mg/kg IV infusion every 2 weeks) or Avastin plus irinotecan (125 mg/m2 IV or 340 mg/m2 IV for patients on enzyme-inducing anti-epileptic drugs) every 2 weeks until disease progression or unacceptable toxicity. The primary endpoints of the study were 6-month progression-free survival (PFS) and objective response rate (ORR) as assessed by an independent review facility (IRF). Other outcome measures were duration of PFS, duration of response and overall survival. Results of the study are summarised in Table 10.

Table 10 Efficacy Results from Study AVF3708g

	Avastin		Avastin + Irinotecan	
Number of patients	85		82	
	Inv	IRF	Inv	IRF
Primary endpoints				
6-month progression-free survival	43.6% (33.0, 54.3)	42.6% (29.6, 55.5)	57.9% (46.6, 69.2)	50.3% (36.8, 63.9)
98% CI (Inv) 97.5% CI (IRF)				

Number of patients	85		82
	Inv	IRF	Inv	IRF
Objective Response Rate 95% CI (Inv) 97.5% CI (IRF)	41.2% (30.6, 53.3)	28.2% (18.5, 40.3)	51.2% (39.9, 62.4)	37.8% (26.5, 50.8)
Secondary endpoints				
Progression-free survival (months)				
Median (95% CI)	4.2 (3.0, 6.9)	4.2 (2.9, 5.8)	6.8 (5.0, 8.2)	5.6 (4.4, 6.2)
Duration of objective responses (months)				
Median (95% CI)	8.1 (5.5,*)	5.6 (3.0, 5.8)	8.3 (5.5,*)	4.3 (4.2,*)
Overall survival (months)				
Median (95% CI)	9.3 (8.2,*)		8.8 (7.8,*)	

ORR was determined using modified McDonald criteria ; Inv = Investigator's assessment; IRF = Independent Review Facility
* Upper limit of the confidence interval could not be obtained

In study AVF3708g, six-month PFS based on IRF assessments was significantly higher (p<0.0001) compared withhistorical controls for both treatment arms: 42.6% in the Avastin arm and 50.3% in the Avastin plus irinotecan armInvestigator assessment: 43.6% in the Avastin arm and 57.9% in the Avastin plus irinotecan arm). Objective responsetaxes were also significantly higher (p<0.0001) compared with historical controls for both treatment arms: 28.2% inthe Avastin arm and 37.8% in the Avastin plus irinotecan arm (Investigator assessment 41.2% in the Avastin arm and51.2% in the Avastin plus irinotecan arm).
The majority of patients who were receiving steroids at baseline and at baseline, including responders and non-responders, were abileto reduce theirsteroid utilization over time while receiving Avastin treatment. The majority of patients experiencing an objective response or prolonged PFS (at week 24) were able to maintain or improve their neurocognitive functions while on study treatment compared to baseline. The majority of patients that remained in the study and were progression free at 24 weeks, had a Karnofsky performance status (KPS) that remained stable.

Epithelial ovarian, fallopian tube and primary peritoneal cancer

Front-line treatment of ovarian cancer

The safety and efficacy of Avastin in the front-line treatment of patients with epithelial ovarian, fallopian tube or primary peritoneal cancer were investigated in three main studies in this setting: GOG-0218, and BO17707 that compared the recurrent setting or prior Avastin treatment, was studied in a phase III trial randomized, double-blind, placebo-controlled trial (AVF4095g). The study compared the effect of adding Avastin to carboplatin and gemcitabine chemotherapy and continuing Avastin as a single agent to progression to carboplatin and gemcitabine alone.
A total of 484 patients with measurable disease were randomized in equal portions to either:

- Carboplatin (AUC4, Day 1) and gemcitabine (1000 mg/m2 on Days 1 and 8) and concurrent placebo every 3 weeks for 6 and up to 10 cycles followed by placebo every 3 weeks (15 mg/kg every 3 weeks) alone until disease progression or unacceptable toxicity
- Carboplatin (AUC4, Day 1) and gemcitabine (1000 mg/m2 on Days 1 and 8) and concurrent Avastin (15 mg/kg Day 1) every 3 weeks for 6 and up to 10 cycles followed by Avastin (15 mg/kg every 3 weeks) alone until disease progression or unacceptable toxicity

The primary endpoint was progression-free survival based on investigator assessment using RECIST criteria. Additional endpoints included objective response, duration of response, safety and overall survival. An independent review of the primary endpoint was also conducted. The results of this study are summarized in Table 14.

Table 14 Efficacy Results from Study AVF4095g

Progression-free survival				
	Investigator assessment*		IRC assessment	
	Placebo + C/G (n = 242)	Bevacizumab + C/G (n = 242)	Placebo + C/G (n = 242)	Bevacizumab + C/G (n = 242)
Median PFS (months)	8.4	12.4	8.6	12.3
Hazard ratio (95%CI)	0.484 [0.388, 0.605]		0.451 [0.351, 0.580]	
p-value	< 0.0001		< 0.0001	
Objective response rate				
	Investigator assessment		IRC assessment	
	Placebo + C/G (n = 242)	Bevacizumab + C/G (n = 242)	Placebo + C/G (n = 242)	Bevacizumab + C/G (n = 242)
% pts with objective response	57.4%	78.5%	53.7%	74.8%
p-value	< 0.0001		< 0.0001	
Overall survival**				
	Placebo + C/G (n = 242)		Bevacizumab + C/G (n = 242)	
Median OS (months)	32.9		33.6	
Hazard ratio (95%CI)	0.952 [0.771, 1.176]			
p-value	0.6479			

* Primary analysis
** Final overall survival analysis performed when approximately 73% of the patients had died

MO22224 (AURELIA)

Study MO22224 evaluated the efficacy and safety of Avastin in combination with chemotherapy for platinum-resistant recurrent ovarian cancer. This study was designed as an open-label, randomized, two-arm Phase III evaluation of Avastin plus chemotherapy (CT+BV) versus chemotherapy alone (CT). A total of 361 patients were enrolled into this study and administered either chemotherapy (paclitaxel, topotecan, or PLD) alone or in combination with Avastin :

- CT Arm (chemotherapy alone):
  - Paclitaxel 80 mg/m2 as a 1-hour IV infusion on Days 1, 8, 15, and 22 every 4 weeks.
  - Topotecan 4 mg/m2 as a 30-minute IV infusion on Days 1, 8, and 15 every 4 weeks. Alternatively, a 1.25 mg/m2 dose could be administered over 30 minutes on Days 1–5 every 3 weeks.
  - PLD 40 mg/m2 as a 1 mg/min IV infusion on Day 1 only every 4 weeks. After Cycle 1, the drug could be delivered as a 1-hour infusion.
- CT+BV Arm (chemotherapy plus Avastin):
  - The chosen chemotherapy was combined with Avastin 10 mg/kg IV every 2 weeks (or Avastin 15 mg/kg every 3 weeks if used in combination with topotecan 1.25 mg/m2 on Days 1–5 on a every 3 weeks schedule).

Eligible patients had ovarian cancer that progressed within 6 months of previous platinum therapy. If a patient had been previously included in a blinded trial with an anti-angiogenic agent, the patient was enrolled in the same platinum or Avastin -containing arm. Patients were stratified according to platinum-free interval and agent.
The primary endpoint was progression-free survival, with secondary endpoints including objective response rate and overall survival. Results are presented in Table 15.

Table 15. Efficacy Results from Study MO22224 (AURELIA)

	Primary Endpoint	
Progression-Free Survival		
	CT (n=182)	CT + BV (n=179)
Median (months)	3.4	6.7
Hazard ratio (95% CI)	0.379 [0.296, 0.485]	
p-value	< 0.0001	
	Secondary Endpoints	
Objective response Rate*		
	CT (n=144)	CT + BV (n=142)
% patients with objective response	18 (12.5%)	40 (28.2%)
p-value	0.0007	
Overall Survival (final analysis)**		
	CT (n=182)	CT + BV (n=179)
Median OS (months)	13.3	16.6
Hazard ratio [95% CI]	0.870 [0.678, 1.116]	
p-value	0.2711	

All analyses presented in this table are stratified analyses.
*Randomized Platinum-Free Interval Disease at Baseline
** At the time of the final OS analysis (25 January 2013), 266 patients (73.7%) had died across the two treatmentarms.

Cervical cancer

GOG-0240

The efficacy and safety of Avastin in combination with chemotherapy (paclitaxel and cisplatin or paclitaxel and topotecan) as a treatment for patients with persistent, recurrent, or metastatic carcinoma of the cervix was evaluated in study GOG-0240, a randomized, four-arm, multi-centre phase III trial.
A total of 452 patients were randomized to receive either:

- Paclitaxel 135 mg/m2 IV over 24 hours on Day 1 and cisplatin 50 mg/m2 IV on Day 2, every 3 weeks (q3w); or paclitaxel 175 mg/m2 IV over 3 hours on Day 1 and cisplatin 50 mg/m2 IV on Day 2 (q3w); or paclitaxel 175 mg/m2 IV over 3 hours on Day 1 and cisplatin 50 mg/m2 IV on Day 1 (q3w).
- Paclitaxel 135 mg/m2 IV over 24 hours on Day 1 and cisplatin 50 mg/m2 IV on Day 2 plus Avastin 15 mg/kg IV Day 2 (q3w); or paclitaxel 175 mg/m2 IV over 3 hours on Day 1 and cisplatin 50 mg/m2 IV on Day 2 plus Avastin 15 mg/kg IV